Kaluza sent his paper to Einstein, and at first Einstein was quite intrigued. On April 21, 1919, Einstein wrote back to Kaluza and told him that it had never occurred to him that unification might be achieved "through a five-dimensional [four space and one time] cylinder-world." He added, "At first glance, I like your idea enormously."4 About a week later, though, Einstein wrote Kaluza again, this time with some skepticism: "I have read through your paper and find it really interesting. Nowhere, so far, can I see an impossibility. On the other hand, I have to admit that the arguments brought forward so far do not appear convincing enough."5 But then, on October 14, 1921, more than two years later, Einstein wrote to Kaluza again, having had time to digest Kaluza's novel approach more fully: "I am having second thoughts about having restrained you from publishing your idea on a unification of gravitation and electricity two years ago…. If you wish, I shall present your paper to the academy after all."6 Belatedly, Kaluza had received the master's stamp of approval.

Although it was a beautiful idea, subsequent detailed study of Kaluza's proposal, augmented by Klein's contributions, showed that it was in serious conflict with experimental data. The simplest attempts to incorporate the electron into the theory predicted relations between its mass and its charge that were vastly different from their measured values. Because there did not seem to be any obvious way of getting around this problem, many of the physicists who had taken notice of Kaluza's idea lost interest. Einstein and others continued, now and then, to dabble with the possibility of extra curled-up dimensions, but it quickly came to be an enterprise on the outskirts of theoretical physics.

In a real sense, Kaluza's idea was way ahead of its time. The 1920s marked the start of a bull market for theoretical and experimental physics concerned with understanding the basic laws of the microworld. Theorists had their hands full as they sought to develop the structure of quantum mechanics and quantum field theory. Experimentalists had the detailed properties of the atom as well as numerous other elementary material constituents to discover. Theory guided experiment and experiment refined theory as physicists pushed forward for half a century, ultimately to reveal the standard model. It is